Atmospheric entry

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use energy and geometry to discuss entry heating without claiming a guaranteed outcome.

SUCCESS CRITERIA - I can...

1. Compare kinetic energy with a stated heating requirement.
2. Explain the role of surface area relative to mass.
3. Name missing factors in a survival prediction.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A system receives 50 J heat and does 30 J work on its surroundings. Find ΔU with $\Delta U = Q - W_{by}$.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $K = \frac{1}{2}mv^2$

$Q_{absorbed}$ = fraction absorbed × energy available.

Entry also involves ablation and fragmentation. An energy comparison alone cannot predict survival.

Worked reasoning: A fictional entering object loses 1000 J of mechanical energy; a model assigns 700 J to heated air and 300 J to the object. Those shares are supplied assumptions, not universal atmospheric-entry fractions.

Your turn: Draw the energy budget and identify one measurement needed to test the shares.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and interpret the model

Compare a small and a large sphere of the same material. Which has more surface area per unit mass? What does that suggest about surface interactions per kilogram?



A B

Marzano check - circle: ① ② ③

04 Apply the physics

An atmospheric-entry simulation assigns a model object 100 J of kinetic energy. A specified heating change requires 20 J. Find the ratio, then draw two possible energy partitions. Does enough total energy prove that all of it heats the object? Explain.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

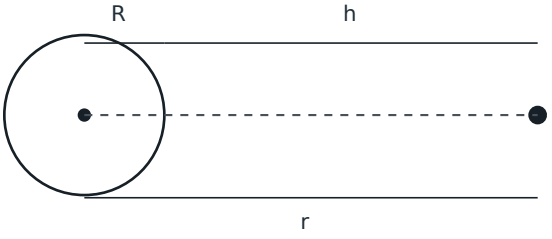
05 Explain your reasoning

Name at least three additional quantities needed before claiming a particular rock completely ablates or reaches the ground.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

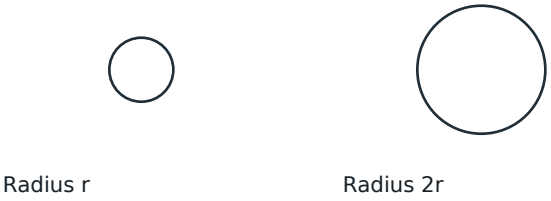
A simplified model requires 24 J to escape, but an object has only 20 J and no later energy input. Does the model predict escape? Explain.
Label the distances or energy levels used in your reasoning.



Marzano check - circle: 1 2 3

07 Exit check

Two equal-density spheres have radii r and 2 r. Supplied fact: area grows 4 times and mass 8 times. Compare area per mass. Label energy entering, leaving or stored on the system model.



Radius r

Radius 2r

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Compare two objects with the same 1000 J energy loss but different assumed fractions heating the object: 20% and 60%. Find both heating amounts and explain why survival is still uncertain 0. Label energy entering, leaving or stored on the system model.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3